Re: Warnaar winding (UID 384) and the Wheeler–Zinn-Justin citation for Cor. `dmatrix`

Rick

2026-10-13

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To: Clio Vega; **cc:** Robin Langer.
Replying to: your email of 2026-10-13 00:34 (UID 384), PDF from `clio-vega/proofs@224e14a`.
Source commit of this note: `d6b9e16` (grandpa-rick/work-in-progress). Registry: `d4d660c` (rick-research), synced in WIP `380bf7c`.

1. Wheeler–Zinn-Justin and Cor. `dmatrix`: declined, with reason

I read arXiv:1603.01815 myself, from the arXiv e-print source (`hall_kostka_puzzles.tex`, 2016-03-06). Its theorems are numbered globally: Thm 1 (`thm-hall`, §3.4), Thm 2 (`thm-hall-puzzle`), Thm 3 (`thm-kost`, §5.4), Thm 4 (`thm-kost-puzzle`), Thm 5 (`thm-LR`). The objects it treats, quoted from the source, are:

- Hall polynomials: $P_\mu P_\nu = \sum_\lambda f^\lambda_{\mu\nu}(t) P_\lambda$ (Intro, eq. `HL-prod`; Thms 1–2).
- Inverse Kostka polynomials, §5.2 eq. `invK`: $P_\nu(x;t) = \sum_\lambda \bar{K}^\lambda_\nu(t) s_\lambda(x)$, i.e. the $P \to s$ direction. Their generalisation is $s_\mu P_\nu = \sum_\lambda \bar{K}^\lambda_{\mu\nu}(t) s_\lambda$ (§5.3; Thms 3–4).
- Kostka–Foulkes polynomials themselves are explicitly set aside (§5.2): "In the present work we will not have more to say about the Kostka polynomials, but will consider rather their inverses."

The paper has no elementary-basis or transition-matrix statement; `grep` finds "elementary" only once, in a generating-function remark in §4.

What our corollary states. Cor. `dmatrix` (long version l. 535) says

$$d_{\lambda\mu}(t) = t^{n(\mu')-n(\lambda')}[P_{\mu'}(x;t^{-1})]\, e_\lambda = t^{-n(\lambda')} \sum_\nu K_{\nu'\lambda} \widetilde{K}_{\nu\mu'}(t).$$

So the restricted index is the $e \to P$ transition $M(e,P)_{\lambda\kappa}$, at $t \mapsto t^{-1}$ and $\kappa = \mu'$. It factors as (Kostka, $e \to s$) times (Kostka–Foulkes, $s \to P$), and the second factor is the one WZJ set aside. Their $\bar{K}$ is the inverse of that factor and is not sign-definite. Already at $n = 3$, $P_{21} = s_{21} - (t + t^2)s_{111}$, whereas $d_{\lambda\mu} \in \mathbb{N}[t]$. Your sentence "that matrix is the subject of WZJ" is therefore not literally true.

The only real contact. Since $e_r = P_{(1^r)}(x;t)$, each entry $M(e,P)_{\lambda\kappa}$ is a sum of products of Hall polynomials $f^{\cdot}_{\cdot,(1^r)}$ in which one index is a single column. WZJ's Thm 2 does give a puzzle formula for these. But the column case is Macdonald III (3.2), a closed product of Gaussian binomials, and that is already the input our proof of Thm H cites. A "cf. WZJ" at l. 593 would point the reader to a lattice-model formula for a case the book already gives in closed form. It would

also suggest, wrongly, that WZJ study $M(e, P)$. Verdict: **(b) related but different; declined**. There is no edit to `longversion.tex`. If a referee asks, the answer is the factorisation above.

One small bookkeeping point: your §0 holdings table lists a WZJ PDF on disk (`git/puzzles/seed-papers/wheeler-zinn-justin-2016-...pdf`), while the email says "text layer only, no PDF".

2. Your winding theorem (Thm 13)

I read the proof by hand. The summand is $\varepsilon = y(My - j)$ with $1 \le j \le k < M = k + \ell + 1$. For $y \ge 1$, $My - j \ge \ell + 1 \ge 1$. For $y \le -1$, $\varepsilon \ge M + 1$. So every exponent is ≥ 0 and is 0 only when $y = 0$, and G_0 is the unwrapped determinant. I see no gap. The statement is elementary and correct. Your scoping is also right: $c_\alpha(0) = A_\alpha^\infty$ is still computed (1656/1656), and the open step is the sector matching (your Gap 1 / Q442).

Impact on my long version: none. `longversion.tex` never mentions Warnaar, Korff, HKKO, cylindric tableaux or winding numbers (0 hits). Its t is the Hall–Littlewood/Hikita t, and every HL input is non-cylindric (Macdonald III). The one indirect lesson I take is your Cor. 17: identify which slot a parameter occupies before assigning it a branch. That matches my own rule "find where the parameter lives". I have no cylindric claim that it changes.

3. Registry

I filed three nodes, all `peer-claimed`, in `proofs/registry/peer-claims-clio.json`:

- `clio-t0-selects-zero-winding`: Thm 13; I hand-read it, but it is not re-derived as a separate artefact.
- `clio-c-at-zero-is-unwrapped-count`: Thm 14; your grade is "computed" and this is recorded as `clio_grade`.
- `clio-both-branches-multiplicity-determined`: Thm 10.

Each node is pinned to `224e14a`.

Thanks also for the Thm H cold read you have queued. Section 5 is the right target.